

SAHIL GALA

Bachelor of Technology| Indian Institute of Technology, Kanpur

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EDUCATION

Indian Institute of Technology, Kanpur
Undergraduate | BTech ME CGPA: 8.70/10
2021 - Present Kanpur, India

Pace Science Junior College
XII, HSC Percentage: 95.33%
2021 Mumbai, India

Gundecha Education Academy
X, ICSE Percentage: 96.8%
2019 Mumbai, India

SKILLS AND UTILITIES

C, C++, Java, JS, Python, DS, Algorithms

Git, Simscape, LaTeX, Matlab, CAD

React JS, Nodejs, Tensorflow, HTML, CSS, ROS

ACHIEVEMENTS & AWARDS

- Secured an **AIR 2682** in JEE Advanced 2021 & **AIR 5181** in JEE Mains 2021 among **1.2 million** candidates.
- Secured **99.88** percentile in Maharashtra Common Entrance Test (CET) .
- Conferred with the **Academic Excellence Award** twice (**2021,2022**) for outstanding academic performance by IIT Kanpur.
- Selected for **SURGE IITK** program for summer research at the end of second year, given to **250** students across India.

COURSEWORK

🎓 : coursera

Computer Science

Unsupervised and supervised Machine Learning by Andrew NG^c
Theory of Computation, Introduction to Programming
Introduction to Back-End development by META^c
Data Structures and Algorithms : Programming Pathshala

Mathematics and Statistics

Partial Differential Equations, Linear Algebra and Diff. Equations
Complex variables analysis, Multivariate Calculus and Real analysis

EXTRA CURRICULARS

Competitive programming

- Maximum rating of 1436 in **codechef**
- Participated in **Google Kickstart 2022** , **IICC** and **Codegaze** (Level 2) competitions.
- Solved 200+ problems on **leetcode**, **Geeks for Geeks** and **Interviewbit**.
- Participated in **codegoda** and **algobash** hackathons , **cisco** ideathon and **flipkart grid challenge**(participating).

Miscellaneous

- Won the inter hall competition **takneek** in domains of **robotics** and **web development** .
- Completed a **marketing** intern at **Geeks for Geeks** promoting its campaigns through social media and spreading **coding** culture via net-working as a part of **GFG community**.

WORK EXPERIENCE

Full Stack Developer at Markoknow

Web Development internship December 2022-February 2023

- Worked on the **frontend** and **backend** of markoknow main website using **MERN** , **Node js** and **Mongo DB** databases.
- Created and worked upon the admin portal of Markoknow using **Express JS** and **mongoose models** and organised the router files via **clean coding**.
- Created a **mail transport system** that could send instantaneous mails of certificates to members of **Markoknow** community.

Undergraduate Research Intern | SURGE IITK

Prof. Shyam Sunder May 2023-July 2023

- Worked on studying **viscous fingering** in porous media by experimenting various liquids on surface of sponge and cloth.
- Implemented image processing in **matlab** to the videos and brought about conclusions regarding fluid behaviour and computed the rate of change of area.
- Performed **PIV cross correlation** analysis between image pairs at different time intervals including correlation computation, and velocity estimation.

PROJECTS

You Look Only Once (YOLO)

Stamatics, IIT Kanpur March 2023- June 2023

- Learned the basics of python and python-libraries such as **numpy** , **pandas**, **seaborn** and **matplotlib** as well as basics of **neural networks**.
- Implemented test models using machine learning algorithms such as **logistic** and **linear** regression and worked on **CNN model** for digit recognition.

Scribble Board

Self Project March 2023 - April 2023

- Implemented a **web application** that allows **freehand scribbling** and plenty of other features using HTML, CSS, Javascript (**Fabric JS**) and Python that provides a powerful, declarative, and event-driven **API** for manipulating and animating canvas elements.
- Incorporated **shape auto completion** feature using **PointNet**- a powerful **deep neural network** architecture and trained auto completion **models**.

Dynamic Darwin- Team Humanoid

Science and Technology Council IITK May 2022-August 2022

- Simulated a **dynamic humanoid robot** for multiple gaits generation using **matlab**, **simulink** and **simscape** and made it perform various bodily movements using **contols**. Worked on **trajectory generation** for individual legs.

POSITIONS OF RESPONSIBILITY

Secretary Web Development

Entrepreneurship Cell ,IITK May 2022-April 2023

- Revamped the website of Policy Page, Bootcamp and CA program of Ecell IITK using **HTML,CSS,Javascript** and **React**.

Secretary, Robotics Club

Science and Technology Council, IITK July 2022-April 2023

- Organised the **SnT pavillion** where we gave fair idea to the freshman about what robotics club does and the projects completed so far.
- Worked upon the **ros** and simulation of **rover** robot that could locomate and transport loads from place to place.

Manager, Research Wing

Academic and Career Council ,IITK April 2023-Ongoing

- Organised **SRC'23**, a 2-day-long multidisciplinary convention on science, technology, humanities, and management, by IIT Kanpur. The convention is the first of its kind and brings together the best college students from **across India**.
- Aided in promoting **research awareness** in the campus by interviewing professors and research scholars, and through various hackathons.